PORTFOLIO PROJECT | SENIOR DATA ANALYST

CustomerIQ

Customer Shopping Behavior Analysis

An end-to-end data analytics project covering exploratory data analysis, SQL-driven business intelligence, customer segmentation, and interactive Power BI dashboarding across 3,900 transactional records.

3,900	18	10	4
Records Analysed	Dataset Columns	SQL Queries	Dashboard Pages

Python	MYSQL	Power BI	Data Analytics
Pandas · SQLAlchemy	Business Intelligence	DAX · Dashboard	EDA · Segmentation

01 — EXECUTIVE SUMMARY

This project delivers a comprehensive, end-to-end analysis of customer shopping behavior using a retail transactions dataset of 3,900 records. Leveraging Python for data preparation, PostgreSQL for business-intelligence queries, and Power BI for interactive dashboarding, this work surfaces actionable insights across revenue distribution, customer segmentation, product performance, discount strategy, and subscription behavior.

3,900	$59.76	3.75	27%
TOTAL CUSTOMERS Unique Transactions	AVG PURCHASE AMOUNT USD Per Transaction	AVG REVIEW RATING Out of 5.0	SUBSCRIPTION RATE 1,053 Subscribers

Key Findings at a Glance

- **Revenue Split:** Male customers generate $157,890 vs. $75,191 for female — a 2.1x gap — suggesting opportunity to deepen engagement with female shoppers.
- **Customer Loyalty:** 80% of customers fall into the Loyal segment (3,116 of 3,900), indicating strong retention but limited new-customer acquisition (only 83 new buyers).
- **Discount Dependency:** Hats lead discount penetration at 50%, followed by Sneakers (49.7%) and Coats (49.1%) — signalling margin risk if discounting continues unchecked.
- **Subscription Gap:** Despite similar average spend ($59.49 vs $59.87), subscribers represent only 27%, leaving significant subscription revenue on the table.
- **Age Group Revenue:** Young Adults drive the highest revenue at $62,143, followed closely by Middle-aged at $59,197 — together contributing over 52% of total revenue.

02 — DATASET & METHODOLOGY

Dataset Overview

Attribute Group	Columns	Description
Customer Demographics	Age, Gender, Location	Profile of each buyer
Subscription & Loyalty	Subscription Status, Previous Purchases	Retention indicators
Purchase Details	Item, Category, Amount (USD), Season, Size, Color	Transaction-level data
Behavior & Engagement	Discount Applied, Frequency, Review Rating, Shipping Type	Post-purchase signals
Engineered Features	age_group, purchase_frequency_days, Purchase_Date	Derived for deeper analysis

Analytical Methodology

Step 1 — Data Ingestion & EDA (Python)

Loaded dataset with pandas. Used df.info() and df.describe() for structural profiling. Identified 37 missing values in Review Rating, imputed using category-level medians to preserve data integrity.

Step 2 — Data Cleaning & Feature Engineering

Standardised column names to snake_case. Engineered age_group (binned ranges), purchase_frequency_days, and a synthetic Purchase_Date column for time-series enablement. Dropped redundant promo_code_used after confirming overlap with discount_applied.

Step 3 — SQL Business Intelligence (PostgreSQL)

Loaded cleaned DataFrame into PostgreSQL via SQLAlchemy. Executed 10 structured business queries covering revenue segmentation, product rankings, discount analysis, and customer loyalty classification using CASE WHEN logic and window functions.

Step 4 — Power BI Dashboard (CustomerIQ)

Built a 4-page interactive Power BI dashboard with DAX measures, cross-filtering slicers (Gender, Category, Subscription Status, Shipping Type), and visualisations including donut charts, bar charts, and KPI cards.

03 — SQL BUSINESS INTELLIGENCE RESULTS

3.1 Revenue by Gender

Gender	Total Revenue (USD)	% of Total Revenue	Avg Transaction
Male	$157,890	67.7%	$60.31
Female	$75,191	32.3%	$58.98

Analyst Insight: Male customers outspend female customers 2.1x in total revenue. However, the per-transaction gap is modest (~$1.33), suggesting volume — not ticket size — drives the disparity. Targeted campaigns for female segments could close this gap significantly.

3.2 Top 5 Products by Average Review Rating

Rank	Product	Avg Rating	Category
#1	Gloves	3.86 / 5.0	Accessories
#2	Sandals	3.84 / 5.0	Footwear
#3	Boots	3.82 / 5.0	Footwear
#4	Hat	3.80 / 5.0	Accessories
#5	Skirt	3.78 / 5.0	Clothing

3.3 Shipping Type vs. Average Purchase Amount

Shipping Type	Avg Purchase Amount (USD)	Interpretation
Express	$60.48	Higher-spend customers self-select premium shipping
Standard	$58.46	Price-sensitive segment; broader customer base
Delta	+$2.02	Modest premium; potential upsell lever

3.4 Subscriber vs. Non-Subscriber Analysis

Status	Total Customers	Avg Spend (USD)	Total Revenue (USD)	% Share
Subscribed (Yes)	1,053	$59.49	$62,645	27%
Non-Subscribed (No)	2,847	$59.87	$170,436	73%

Analyst Insight: Subscribers and non-subscribers spend virtually the same per transaction. The 73% non-subscriber majority represents significant untapped subscription revenue — loyalty perks or exclusive discounts could convert a meaningful portion of these customers.

3.5 Discount-Dependent Products

Rank	Product	Discount Rate (%)	Risk Assessment
#1	Hat	50.00%	HIGH — 1 in 2 sales requires a discount
#2	Sneakers	49.66%	HIGH — approaching discount dependency
#3	Coat	49.07%	HIGH — seasonal markdown risk
#4	Sweater	48.17%	MEDIUM-HIGH — winter clearance pattern
#5	Pants	47.37%	MEDIUM-HIGH — monitor closely

04 — CUSTOMER SEGMENTATION ANALYSIS

Customers were classified into three behavioural segments based on previous purchase count, enabling tailored retention and acquisition strategies.

Segment	Definition	Count	% Share	Strategic Focus
Loyal	Previous purchases >= 10	3,116	79.9%	Reward & Upsell
Returning	Previous purchases 2 to 9	701	18.0%	Nurture & Upgrade
New	Previous purchases 0 to 1	83	2.1%	Onboard & Convert

Segmentation Insights

- The 80% Loyal base is a major competitive strength — focus should be on maintaining satisfaction and increasing average order value through personalised recommendations.
- The Returning segment (18%) sits at a critical conversion point. Targeted win-back campaigns and loyalty incentives can efficiently migrate these customers to Loyal status.
- The New segment at 2.1% (83 customers) indicates limited new-customer acquisition — investment in awareness campaigns and referral programs is strongly advisable.
- Repeat buyers with >5 purchases are 72.5% non-subscribers (2,518 vs 958), revealing a major opportunity to convert engaged, frequent buyers into paid subscribers.

Revenue by Age Group

Age Group	Total Revenue (USD)	% of Total	Strategic Priority
Young Adult (18-35)	$62,143	26.6%	Primary — digital-first campaigns
Middle-Aged (36-55)	$59,197	25.4%	High-value — loyalty and premium upsell
Adult (upper 36-55)	$55,978	24.0%	Cross-sell outerwear and accessories
Senior (56+)	$55,763	23.9%	Service-led retention focus

Analyst Insight: Revenue is remarkably balanced across all four age groups, ranging from $55,763 to $62,143. This resilience reduces concentration risk but also suggests a one-size-fits-all approach — differentiated age-based messaging could unlock incremental revenue.

05 — PRODUCT PERFORMANCE ANALYSIS

Top 3 Products per Category (by Total Orders)

Rank	Category	Product	Total Orders
#1	Accessories	Jewelry	171
#2	Accessories	Sunglasses	161
#3	Accessories	Belt	161
#1	Clothing	Blouse	171
#2	Clothing	Pants	171
#3	Clothing	Shirt	169
#1	Footwear	Sandals	160
#2	Footwear	Shoes	150
#3	Footwear	Sneakers	145
#1	Outerwear	Jacket	163
#2	Outerwear	Coat	161
#3	Outerwear	Hoodie	159

Analyst Insight: Order volumes are tightly clustered (145-171) across all categories, indicating a balanced product portfolio. Jewelry, Blouses, Pants, Sandals, and Jackets are respective category anchors — prime candidates for featured placement in marketing campaigns.

High-Spending Discount Users

839 customers (21.5% of the base) applied a discount yet still spent above the dataset average of $59.76. This segment represents high-value, deal-responsive buyers — a strategically important cohort for targeted promotions that preserve margin while maintaining engagement.

Metric	Value	Interpretation
Qualifying Customers	839	Used discount AND spent above average
% of Total Base	21.5%	Significant premium deal-seeker segment
Min Purchase in Group	$62	All members well above $59.76 baseline
Max Purchase in Group	$100	High-ceiling spenders attracted by discounts

06 — POWER BI DASHBOARD: CUSTOMERIQ

The CustomerIQ Power BI dashboard translates all analytical findings into an interactive, stakeholder-ready reporting tool. The dashboard is structured across four thematic pages, each serving a distinct analytical audience.

Dashboard Page	Key Visuals	Primary Audience
Page 1 — Executive Overview	KPI cards, Donut (Subscription %), Revenue/Sales by Category	C-Suite / Senior Management
Page 2 — Sales Deep Dive	Revenue by Category, Sales by Age Group, Time-series trend	Sales Director / Category Mgr
Page 3 — Customer Segmentation	Segment distribution, Age group heatmap, Gender revenue split	CRM / Marketing Team
Page 4 — Product & Shipping	Top products, Discount-dependent items, Shipping vs. spend	Product Manager / Ops Team

DAX Measures Implemented

DAX Measure Name	Formula / Logic
Total Revenue	SUM(shopping[purchase_amount_usd])
Avg Purchase Amount	AVERAGE(shopping[purchase_amount_usd])
Avg Review Rating	AVERAGE(shopping[review_rating])
Subscription Rate %	DIVIDE(COUNTROWS(FILTER(...Yes...)), COUNTROWS(shopping))
High-Spend Discount Customers	COUNTROWS(FILTER(... discount=Yes AND amount > [Avg Purchase Amount]))

07 — STRATEGIC BUSINESS RECOMMENDATIONS

1. Scale Subscription Acquisition

With 73% of customers unsubscribed — despite identical spending patterns — launch a value-led subscription campaign. Offer early access to new collections, free express shipping, and birthday discounts to incentivise sign-ups. Target the 2,518 frequent buyers who are not yet subscribers as the highest-priority conversion cohort.

2. Protect Margins on Discount Products

Hat, Sneakers, and Coat products each have ~50% discount rates. Review the promotion calendar — shift from blanket discounts to personalised, threshold-based offers (e.g., Spend $80, get 15% off Hats) to protect gross margin while maintaining conversion rates.

3. Targeted Female-Segment Campaigns

Female customers contribute 32% of revenue vs. 68% for males, despite comparable average transaction values. The gap is volume-driven. Invest in category-specific campaigns for Clothing and Accessories through social commerce and influencer partnerships.

4. Returning-to-Loyal Conversion

The 701 Returning customers (18% of base) represent the clearest near-term loyalty opportunity. A structured nurture sequence — personalised email with product recommendations and a milestone reward at purchase #10 — could convert a meaningful portion to Loyal status within 6-12 months.

5. Leverage Express Shipping as Upsell

Express shipping customers average $2.02 more per transaction. Introduce an Express Perks bundle combining priority shipping with a small discount on repeat purchases — converting the $2 signal into a $5-$8 incremental revenue opportunity per engaged customer.

6. New Customer Acquisition Investment

Only 83 customers (2.1%) are classified as New — a critical warning signal. Allocate budget toward paid search for category-specific terms, referral incentives for Loyal customers, and abandoned-cart remarketing. Set a KPI target to grow the New segment to >=5% within 12 months.

08 — CONCLUSIONS & NEXT STEPS

This analysis demonstrates the power of integrating Python, SQL, and Power BI into a cohesive analytics workflow. The CustomerIQ project moves beyond descriptive statistics to deliver prescriptive, commercially grounded recommendations that directly support business growth.

Analytical Achievements

- Cleaned and engineered a production-quality dataset with feature creation (age bins, synthetic dates) and robust null imputation via category-level medians.
- Designed and executed 10 SQL business queries in PostgreSQL covering revenue attribution, cohort segmentation, product ranking (window functions), and discount analytics.
- Built a 4-page Power BI dashboard with 5+ custom DAX measures, interactive cross-filtering, and stakeholder-facing KPI layout.
- Identified $108,000+ in untapped subscription revenue potential by analysing the non-subscriber frequent-buyer segment.
- Quantified discount risk across 5 high-penetration products, enabling a data-driven pricing review.

Recommended Next Steps

Horizon	Action Items
Short-Term (0-3 months)	Deploy subscription conversion campaign targeting 2,518 non-subscriber frequent buyers. Implement personalised discount thresholds for Hat, Sneakers, Coat.
Medium-Term (3-6 months)	Build a time-series revenue model using the Purchase_Date column. Integrate real-time data refresh in Power BI via API or PostgreSQL direct query.
Long-Term (6-12 months)	Develop a predictive churn model (logistic regression / random forest) on the Returning segment. Apply RFM scoring for dynamic segmentation in CRM.